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GEOM 4008 Final Project Report

Kriging Interpolation Comparison on Alberta Weather Station Elevation Data using System for Automated Geoscientific Analysis (SAGA GIS)

Link

December 21st, 2024

Purpose of the tutorial, and approach taken

The purpose of this tutorial is to introduce users to SAGA GIS, a software that is less commonly known, and less taught at Carleton, but offers powerful geospatial analysis capabilities. The focus is on exploring the intricacies of the kriging interpolation technique, demonstrating how it can be modified in various ways to suit specific analysis needs. This includes examining how changes in input points influence the output, as well as understanding the mathematical functions and variables available for customization. By doing so, the tutorial aims to provide users with a deeper understanding of kriging and its practical applications in geospatial data analysis.

The tutorial incorporates a selection of tools and methods. During the data preparation stage, significant preprocessing was required to get the data ready for use; however, these steps were omitted from the tutorial itself because they were not its primary focus, and the target audience would already be familiar with such tasks. For instance, to create a limited dataset of weather stations, the first 20 stations were selected alphabetically after a quick check to ensure they were reasonably distributed geographically. While no Alberta-specific shapefile was available, a Canadian shapefile was clipped to isolate the province, providing a suitable boundary for focused analysis. The kriging tool in SAGA GIS was the sole interpolation method demonstrated, keeping the tutorial concise. Additionally, error calculations and comparisons were emphasized, using RMSE values and side-by-side map visualizations to facilitate meaningful analysis.

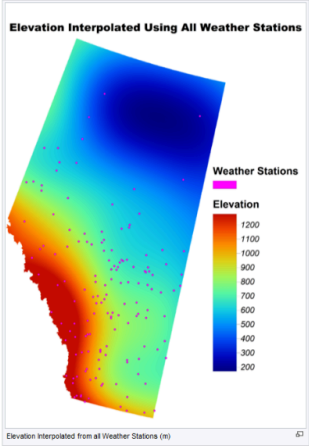
The tutorial is designed for an audience familiar with the fundamentals of geomatics but who may not yet understand kriging in detail. It targets students in post-secondary education and early-career professionals who are seeking to expand their knowledge of interpolation methods. Recognizing that these users may be unfamiliar with SAGA GIS, the tutorial adopts a

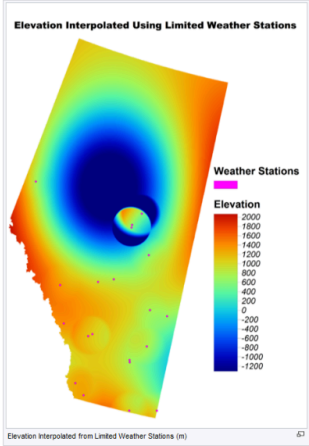
detailed and step-by-step approach, ensuring clarity and accessibility. Special attention is given to making navigation within the software straightforward, with clear instructions on where to click and how to proceed. This approach minimizes potential frustration and enables users to focus on the learning objectives. By tailoring the tutorial in this way, it effectively meets the needs of its target audience and achieves its goal of demystifying both the software and the kriging technique.

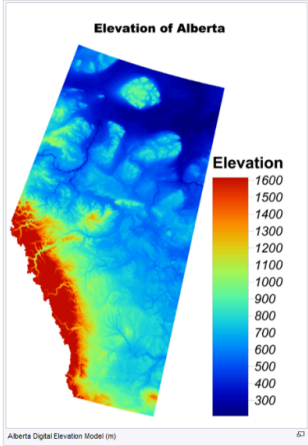
Developing this tutorial has been quite the educational process, offering valuable insights into both the technical aspects of SAGA GIS and broader concepts in geomatics. One of the challenges we faced was the difficulty of finding suitable open data, underscoring the importance of reliable data sources for geospatial analysis. Through this project, we also learned how SAGA GIS operates, exploring its unique strengths and limitations compared to more commonly used software like QGIS or ArcGIS. Additionally, we gained practical experience with Carleton Library's GIS services and learned all the Wiki-based content formatting, which was essential for presenting the tutorial effectively.

When it came to Wiki formatting, I found that borrowing the structure and formatting style from an existing tutorial was the most effective approach. This method provided a solid foundation to work from, starting with background information and a brief introduction to set the context of the tutorial. The instructions were then broken into chapters and steps, making the content easy to follow and visually digestible. Working in a pair was particularly beneficial, as it allowed us to divide tasks efficiently. For example, Alex handled uploading all the tutorial images, a time-intensive task that was necessary to the final product. Formatting the images was one of my tasks and that required many “preview changes” to get right, especially in the “Visual Comparisons” section when we needed 3 images the same size beside one another.

Visual Comparisons





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===Visual Comparisons===
<div style="display: flex; justify-content: center; gap: 10px">

  <div style="text-align: center;">
    [[Image:SAGAfullmap.png|thumb|none|frame|Elevation Interpolated from all Weather Stations (m)|450px]]
  </div>
  <div style="text-align: center;">
    [[Image:SAGAlimitedmap.png|thumb|none|frame|Elevation Interpolated from Limited Weather Stations (m)|450px]]
  </div>
  <div style="text-align: center;">
    [[Image:AlbertaDEM2.png|thumb|none|frame|Alberta Digital Elevation Model (m)|450px]]
  </div>

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Discussion and conclusions

Our background learning in this course played a significant role in shaping the tutorial's development. In Week 2, we examined various types of error classification, including RMSE, which became our preferred metric for evaluating interpolation accuracy in the tutorial. Week 3 focused on open data and data storage, emphasizing the importance of using openly available resources. This concept informed our decision to ensure that all data and software used in the project adhered to open-access principles. The discussions on open data also highlighted the benefits of sharing data for the geomatics community, which inspired us to transition from a temporary Google Drive link to hosting the data on Carleton's servers—a task we plan to complete after the holidays.

Regarding the tools used in the project, SAGA GIS proved to be both a strength and a limitation. Firstly, the installation of the software was quite easy with no fancy installers needed. While its kriging tool was robust and well-suited for our analysis, the software's documentation, particularly on variograms, lacked depth. Variograms are integral to kriging as they define spatial relationships in the data, yet the documentation provided insufficient guidance on their usage. In contrast, the layout tool in SAGA, while initially less intuitive than those in mainstream software, ultimately offered a streamlined experience once mastered. Compared to QGIS and ArcGIS, SAGA GIS does have fewer features overall, which limits its flexibility but can make it more focused for specific tasks like kriging.

The conclusion of our tutorial underscored some key findings. We successfully demonstrated that using a greater number of input points leads to a more accurate kriging interpolation. However, we encountered mixed results when it came to variogram selection. For instance, while the Gaussian variogram produced visually unsatisfactory results for the limited weather station dataset, its statistical determination was superior, which ultimately influenced our choice. In contrast, the linear variogram yielded more visually appealing maps but did not perform as well by our chosen metric. These trade-offs highlight the complexities of kriging and the importance of carefully selecting parameters based on the specific goals of the analysis.

Looking ahead, there are several opportunities for improving the tutorial. First, we omitted instructions on how to compare the interpolation results, opting instead to only present our conclusions. Future versions of the tutorial could include a step-by-step guide for performing these comparisons, encouraging users to conduct their own analyses and potentially discover more accurate kriging configurations. Second, while we touched on variograms briefly, expanding this section to explore their impact on kriging outcomes would add significant value. Lastly, selecting a study area with more pronounced elevation variability could yield clearer

insights into interpolation differences. Alberta's terrain, while suitable for this project, includes a stark contrast between the Rockies and the flatlands, which may not be ideal for demonstrating subtler differences in kriging techniques.

In conclusion, this tutorial development highlighted the challenges and opportunities of working with open data and open-source software, while reinforcing the importance of concepts like uncertainty, error classification, and community standards in geomatics. SAGA GIS's tools, though limited in some respects, provided an effective platform for demonstrating kriging. Future enhancements to the tutorial could focus on expanding user engagement, improving documentation, and exploring alternative study areas to maximize its educational potential.